Operator Precedence

Operator precedence determines the order in which the operators in an expression are evaluated.

For eg –

*int x = 3 * 4 - 1;*

In the above example, the value of x will be 11, not 9. This happens because the precedence of * operator is higher than - operator. That is why the expression is evaluated as (3 * 4) - 1 and not 3 * (4 - 1).

Operator Precedence Table

Operators	Precedence
postfix increment and decrement	++ --
prefix increment and decrement, and unary	++ -- + - ~ !
multiplicative	* / %
additive	+ -
shift	<< >> >>>
relational	< > <= >= instanceof
equality	== !=
bitwise AND	&
bitwise exclusive OR	^
bitwise inclusive OR	|
logical AND	&&
logical OR	||
ternary	? :
assignment	= += -= *= /= %= &= ^= |= <<= >>= >>>=

Associativity of Operators

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If an expression has two operators with similar precedence, the expression is evaluated according to its **associativity** (either left to right, or right to left).

Operators	Precedence	Associativity
postfix increment and decrement	`++` `--`	left to right
prefix increment and decrement, and unary	`++` `--` `+` `-` `~` `!`	right to left
multiplicative	`*` `/` `%`	left to right
additive	`+` `-`	left to right
shift	`<<` `>>` `>>>`	left to right
relational	`<` `>` `<=` `>=` `instanceof`	left to right
equality	`==` `!=`	left to right
bitwise AND	`&`	left to right
bitwise exclusive OR	`^`	left to right
bitwise inclusive OR	`|`	left to right
logical AND	`&&`	left to right
logical OR	`||`	left to right
ternary	`?:`	right to left
assignment	`=` `+=` `-=` `*=` `/=` `%=` `&=` `^=` `|=` `<<=` `>>=` `>>>=`	right to left

Note - These notes are just for a quick glance. We don't have to memorize them all at once. Most of these rules are very logical and we have been following them in a lot of instances already.

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